

WADDINGTON, United Kingdom

WMO: 033770

Lat: 53.1753N

Lon: 0.5233W

Elev: 231

StdP: 14.57

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.0	29.0	21.4	16.2	31.8	23.6	18.0	32.6	32.2	48.3	29.6	47.9	7.9	330	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	78.9	64.4	75.3	62.8	72.2	61.5	66.3	75.0	64.5	72.4	63.0	70.1	10.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	87.0	69.3	61.5	82.2	67.5	60.1	78.1	66.0	31.0	75.2	29.7	72.2	28.6	70.4	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.9	23.7	21.1	DB	22.4	85.6	3.7	3.0	19.7	87.8	17.6	89.5	15.5	91.2	12.9	93.3	(4)
(5)			WB	21.5	69.6	3.5	2.1	19.0	71.1	17.0	72.4	15.0	73.6	12.4	75.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	39.8	40.6	43.6	47.8	53.0	58.4	62.6	62.4	57.7	51.8	44.8	40.4
	DBStd	9.69	5.42	5.72	5.39	5.25	5.22	4.69	4.59	4.34	4.16	4.96	5.29	6.22
	HDD50	1428	318	266	207	102	27	1	0	0	1	37	169	300
	HDD65	5434	782	683	663	516	371	206	104	107	223	411	607	761
	CDD50	1536	1	3	9	36	121	251	391	383	232	92	13	4
	CDD65	66	0	0	0	0	0	7	30	25	3	1	0	0
	CDH74	445	0	0	0	1	6	64	201	149	21	3	0	0
	CDH80	69	0	0	0	0	0	9	31	28	1	0	0	0
Wind	WSAvg	10.5	11.9	11.9	11.4	10.6	10.3	9.3	9.0	9.2	9.7	10.9	10.9	11.5
Precipitation	PrecAvg	23.7	2.0	1.5	1.7	1.8	1.9	2.1	2.2	2.4	1.9	2.1	2.2	2.0
	PrecMax	32.4	4.6	5.2	4.3	6.1	4.6	7.9	5.2	5.7	5.4	5.2	4.7	6.4
	PrecMin	15.1	0.5	0.1	0.2	0.3	0.2	0.2	0.5	0.2	0.3	0.4	0.7	0.6
	PrecStd	3.8	0.9	0.9	0.9	1.2	1.1	1.5	1.3	1.1	1.2	1.2	0.9	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.0	56.8	60.8	69.4	74.2	81.1	83.9	83.7	77.2	68.4	60.1	56.6
		MCWB	51.6	51.6	51.8	56.4	61.5	66.3	66.5	65.4	62.5	60.7	56.5	52.8
	2%	DB	52.4	53.6	57.3	63.6	69.9	74.7	79.0	78.4	71.8	63.9	57.0	54.3
		MCWB	49.2	49.5	50.2	53.6	59.0	62.2	64.6	64.5	61.7	58.0	53.9	51.5
	5%	DB	50.2	51.2	54.6	59.9	66.2	70.7	75.6	73.7	68.1	61.4	54.8	51.9
		MCWB	47.5	47.5	49.1	51.6	56.6	60.3	63.2	62.6	59.6	56.8	51.8	49.4
	10%	DB	48.0	49.2	52.3	56.9	62.8	67.5	72.1	70.8	65.2	59.2	52.6	49.6
		MCWB	45.5	45.8	47.3	50.1	54.6	58.6	61.7	60.9	57.9	55.2	50.1	47.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.3	52.6	53.9	58.0	63.1	67.9	68.7	68.2	65.5	61.8	57.2	53.6
		MCDB	54.8	56.0	58.2	66.4	71.9	78.3	79.9	77.4	72.6	67.0	59.5	55.6
	2%	WB	49.7	50.0	51.7	55.0	60.0	63.9	66.3	65.9	63.0	59.5	54.5	51.9
		MCDB	51.8	53.1	55.7	61.6	68.0	71.4	75.5	74.1	69.3	62.9	56.5	54.1
	5%	WB	47.9	48.1	49.8	52.8	57.8	61.7	64.5	64.2	61.0	57.6	52.3	49.7
		MCDB	49.8	50.8	54.0	58.7	64.4	68.4	72.5	71.9	65.8	60.6	54.2	51.7
	10%	WB	45.8	46.2	48.0	50.7	55.8	59.9	62.9	62.6	59.4	55.9	50.4	47.5
		MCDB	47.8	48.8	51.8	56.1	61.4	66.2	70.6	69.0	63.7	58.6	52.2	49.5
Mean Daily Temperature Range	5% DB	MDBR	8.3	9.7	11.8	13.9	14.6	14.5	15.4	14.7	13.3	11.2	9.2	8.1
		MCDBR	10.8	11.3	14.8	18.4	19.9	20.3	21.4	19.2	17.5	12.8	10.5	10.3
		MCWBR	9.9	8.9	9.5	10.4	11.0	10.3	9.4	8.3	9.2	8.5	8.9	9.4
	5% WB	MCDBR	10.8	11.0	13.6	17.4	18.4	18.2	19.0	17.3	15.1	12.4	10.3	10.1
		MCWBR	10.1	9.2	9.8	10.6	11.0	10.4	9.5	8.3	9.4	9.3	9.1	9.5
Clear-Sky Solar Irradiance	taub	0.322	0.338	0.370	0.396	0.397	0.403	0.404	0.391	0.370	0.350	0.324	0.310	
	taud	2.359	2.391	2.298	2.236	2.274	2.281	2.288	2.347	2.411	2.446	2.410	2.334	
	Ebn at Noon	207	242	256	264	270	269	266	263	255	235	205	187	
	Edh at Noon	19	24	32	39	40	40	39	35	29	23	18	16	
All-Sky Solar Radiation	RadAvg	220	437	777	1198	1498	1565	1519	1251	927	533	272	178	
	RadStd	21	51	94	141	98	150	161	84	84	46	30	18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page